GCReW Flume and Tidal Creek Bathymetry Survey

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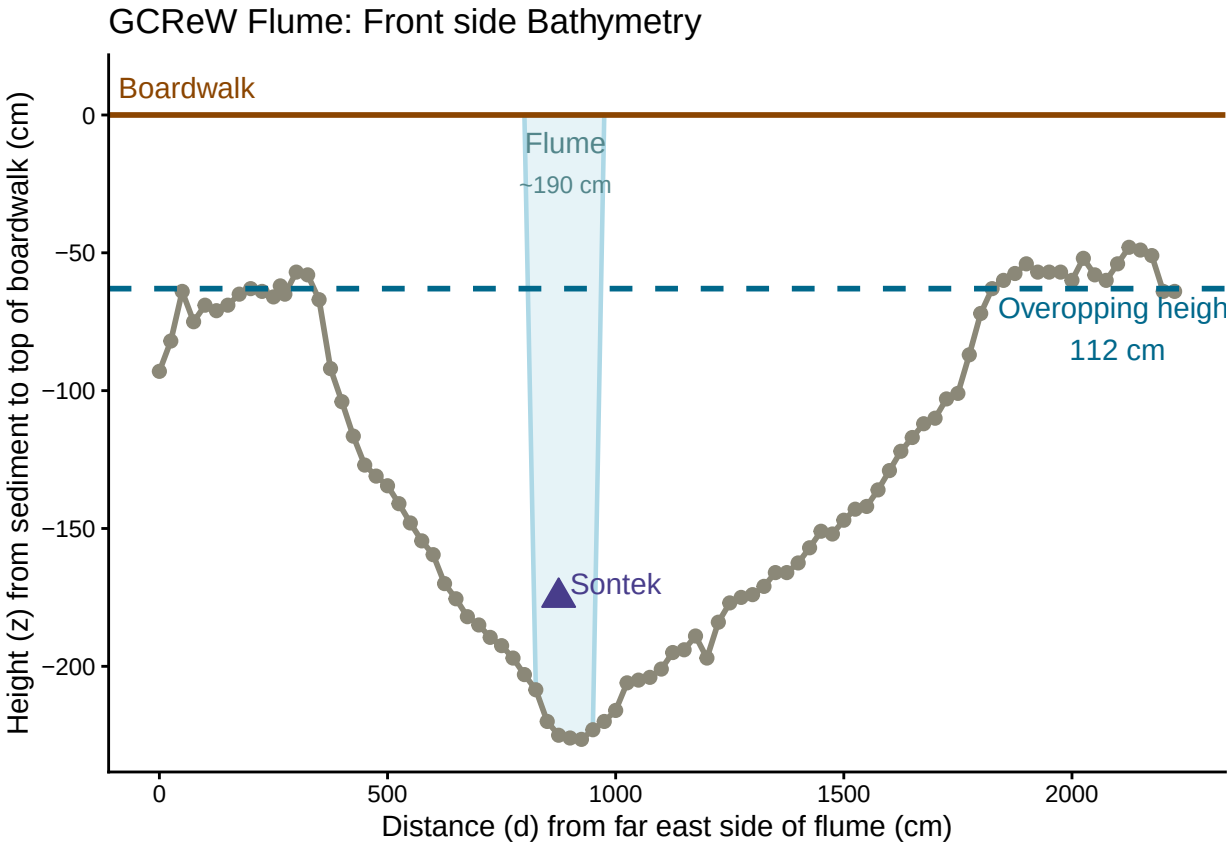
Bathymetry Measurements:

The height (z) from the sediment surface to the top of the boardwalk was measured every 25cm from the east side of the flume, across the tidal creek, to the other end (toward CO2 lab). We measured creek bathymetry along the flume front side (facing out to Rhode River) and back side (facing the organ). The flume opening is ~191.5cm wide. The Sontek IQ ADCP from sensors to bolt (176cm) and bolt to boardwalk (1cm) were also measured. Therefore, the height of tidal creek banks (63cm relative to the boardwalk) and the water depths measured by the Sontek IQ (ADCP) can be used to determine a depth at which overtopping would occur which would be ~112cm water heights.



Figure 1: Arial photo of the GCReW Tidal Creek and Flume

Figures of tidal creek bathymetry, sontek location, flume width and overtopping height:



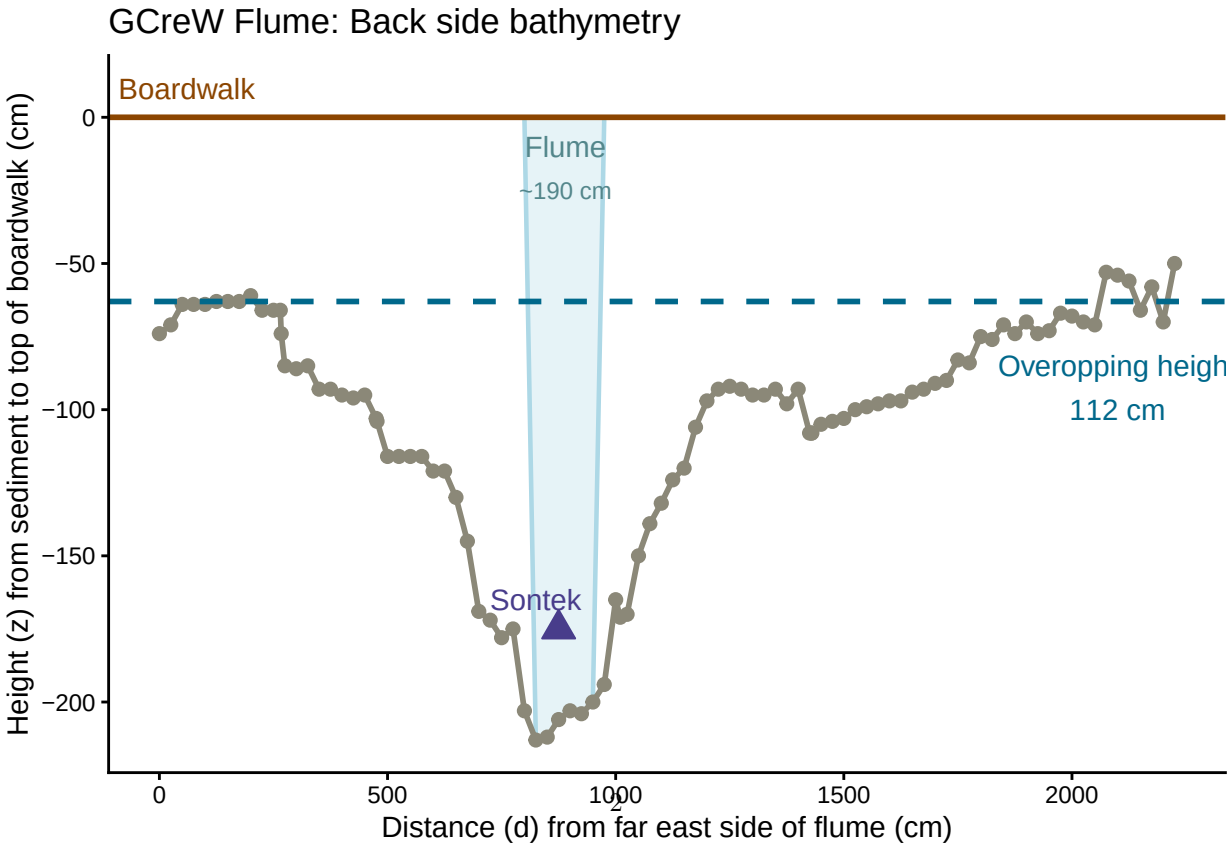




Figure 2: Back side of GCReW Tidal Creek and Flume



Figure 3: Measuring Sontek from sensor to bolt (176 cm)